SAP JOULE ENABLEMENT

Runbook: getting a Joule-enabled SAP environment up and running

Covers SAP-managed and customer-managed provisioning, SAP BTP and identity foundations, every major SAP product integration, Joule Studio, document grounding, Microsoft 365 Copilot, testing, operations and troubleshooting.

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1. How to use this runbook

Joule is SAP's generative-AI assistant. It sits on SAP Business Technology Platform (BTP), signs users in through SAP Cloud Identity Services, finds the right apps through SAP Build Work Zone, and acts in each connected SAP product with that user's own permissions.

There are two ways to switch it on, and the right one depends on your products and how much control you need. Read sections 2 to 5 first; they apply to everyone. Then follow either section 6 (SAP-managed) or section 7 (customer-managed), and only the product chapters in section 8 that you own.

1.1 Reading path

If you are...	Read
Deciding whether and how to start	2 Choose your path · 3 Licences · 4 Landscape design
Starting fresh with SAP cloud products	5 People and access · 6 SAP-managed provisioning · 11 Testing
Setting it up yourself on SAP BTP	5 · 7 Common foundation · the product chapters in 8 · 11 · 12
Adding custom skills, documents or Microsoft 365	9 Extending Joule
Enabling consultants or ABAP developers	10 Joule for Consultants and Joule for Developers
Fixing a problem	13 Troubleshooting · Appendix B SAP Notes
Preparing go-live	14 Security · 15 Master go-live checklist

1.2 Conventions

- **Bold** marks names you click or select in a cockpit or app. `Monospace` marks technical names, IDs, paths and URLs.
- The amber tag VERIFY marks a detail that came from a single or non-SAP source, or where sources disagree. Check it against the SAP Help page named in the step before you rely on it.
- SAP releases Joule changes frequently (SAP Help pages for Joule were updated through October 2026). Treat SAP Help and the SAP Notes in Appendix B as the final word.

> **Why the verification tags**
> This runbook was compiled from search extracts of SAP Help, SAP Learning, the SAP Architecture Center and SAP Community, because direct access to those sites was blocked in the research environment. Plan names, role names and menu paths change between releases, so confirm the tagged items during your own setup.

2. Choose your path

2.1 The two provisioning models

	SAP-managed provisioning	Customer-managed provisioning
What it is	You request Joule in SAP for Me. SAP sets up BTP, identity and the integrations, and runs lifecycle operations.	You set up the BTP subaccount, SAP Cloud Identity Services trust, SAP Build Work Zone and run the Joule booster yourself.
Status	Announced at SAP Sapphire 2026 as the standard activation path for new Joule deployments.	Fully supported for existing deployments; SAP offers migration help towards SAP-managed.
Effort	Low: guided steps in SAP for Me, then product-side settings.	Medium to high: BTP, identity, Work Zone, destinations, roles.
Control	Less: SAP owns the BTP side.	Full: your own subaccount, roles, destinations, extensions.
Best for	New Joule customers on SAP cloud products who want the fastest route.	Landscapes that need Joule Studio, document grounding, Microsoft 365 Copilot, S/4HANA Private Edition specifics, or full control over BTP.
Where	Section 6	Section 7, then section 8

> **Mixing models**
> SAP documents that running both models in parallel is possible, but recommends one model per landscape for a consistent experience (stated for Joule Studio). Decide early, because the BTP subaccount layout differs.

2.2 Path by product

Product	Typical path	Key extra requirement	Chapter
S/4HANA Cloud Public Edition	SAP-managed or booster	Communication arrangements created by the booster; Content Security Policy	8.1
S/4HANA Cloud Private Edition (RISE)	Customer-managed (booster or Cloud Integration Automation service)	Release 2021 or later (2023 FPS01 recommended), Cloud Connector, principal propagation, content exposure	8.2
S/4HANA on-premise / ECC	Hybrid, subject to SAP eligibility	Cloud commitment conditions announced 2026 VERIFY	8.3
SAP SuccessFactors	SAP-managed or booster	Role-based permission "Access to Joule"	8.4
SAP Concur	Booster	Concur domain in security settings	8.5
SAP Ariba	Booster	Ariba OAuth application, realm data centre	8.6
SAP Analytics Cloud	Booster	Work Zone mandatory since Joule 2506; "Just Ask" enabled	8.7
SAP Datasphere	Booster	Role DW Joule Consumer; OAuth client	8.8

Product	Typical path	Key extra requirement	Chapter
SAP IBP	Booster	Catalog SAP_CORE_BC_UI_JOU_PC	8.9
Signavio, Fieldglass, IPD, CX and others	Per SAP Help integration page	See each product guide	8.10

3. Licences and commercial prerequisites

> **Confirm with your SAP account team**
> Commercial terms for Joule changed in July 2025 and again in 2026. Use this section to prepare the conversation, not as a price list.

3.1 What you need

Item	What it covers	How to check
Licence for each SAP product	Joule only works in products you license (for example S/4HANA Cloud, SuccessFactors, Concur).	SAP for Me → Portfolio & Products
Joule Base	Included with qualifying SAP cloud subscriptions. Covers the core assistant (navigation, informational and transactional scenarios).	SAP for Me → Portfolio & Products → Business AI
AI Units (Joule Premium)	Consumption currency for premium capabilities such as Joule Agents and assistants. Purchased and activated in SAP for Me; unused units may expire at the end of the term. VERIFY	SAP for Me; order form
Onboarding SKU	SAP's onboarding guide lists "purchased the AI Unit SKU `8019164` or Joule Base" as a prerequisite. VERIFY	Order form
SAP BTP enterprise global account	Needed for customer-managed provisioning, Joule Studio, document grounding. Contract types cited: CPEA, BTPEA or subscription.	BTP cockpit → global account
SAP Build licence (Joule Studio)	SAP Build entitlement plus SAP Build developer licence to build, test and deploy skills and agents; Joule entitlement to run them in production.	BTP entitlements
Microsoft 365 Copilot licence	Only for the Microsoft 365 Copilot / Teams integration where Copilot skills are used.	Microsoft 365 admin centre

3.2 Questions to settle with SAP

- Which of our products are entitled to Joule Base today, and in which data centres?
- How many AI Units do we need for the agents we plan, and when do they expire?
- Does our contract allow SAP-managed provisioning, or do we stay customer-managed?
- For on-premise S/4HANA or ECC: are we eligible for the 2026 hybrid Joule offering, and under which cloud commitment? VERIFY
- Are there current promotions (for example the 2026 free run of custom Joule agents for SAP Build customers)?

4. Landscape and architecture design

4.1 How Joule fits together

Component	Role in Joule
Joule (BTP service, technical name `das-application`)	The assistant itself: dialogue, intent recognition, generative answers, scenario routing.
SAP Cloud Identity Services: Identity Authentication (IAS)	Single sign-on for Joule and every connected product. One IAS tenant per landscape tier.
SAP Cloud Identity Services: Identity Provisioning (IPS)	Synchronises users, groups and roles (SCIM 2.0, including `userUuid`) to Work Zone and products.
SAP Build Work Zone, standard edition	Navigation service: resolves "open the app for..." requests to the right app the user is allowed to use.
Formation (BTP System Landscape)	Groups Joule with the product systems it integrates; created by the booster.
Destinations	How Joule and Work Zone reach each product system.
Cloud Connector	Secure outbound tunnel to private systems (S/4HANA Private Edition, on-premise).
Generative AI hub in SAP AI Core	SAP-managed model routing behind Joule; not configured by customers.

4.2 Design rules

- **One IAS tenant per tier.** Joule and every product integrated with it must authenticate against the same SAP Cloud Identity Services tenant. Different IAS tenants mean separate Joule instances.
- **One Global User ID per person.** Each user must carry the same Global User ID (`user_uuid`) in every connected product. This is the most common cause of failed sessions.
- **Two tiers.** SAP's landscape recommendation is a non-production Joule (on the **Joule Preview** landscape) for sandbox, development and test, and a production Joule. Older guidance shows three stages.
- **Same subaccount.** Joule and SAP Build Work Zone, standard edition live in the same subaccount, with the Work Zone destinations. Joule talks to only one Work Zone instance; SAP recommends reusing an existing Work Zone.
- **Same region.** Keep the subaccount, IAS tenant and Work Zone in the same region, with the Cloud Foundry environment enabled.
- **Supported regions differ by product.** Examples seen: Joule for Consultants EU10, EU11, EU20, US10, AP10, JP10; Joule Studio AP10, EU10, EU30, EU20, US10, US21; Joule Preview entitlements only on EU10 or EU30. Check the SAP Discovery Center before creating subaccounts. VERIFY
- **Separate subaccounts where SAP requires them.** Joule for Consultants needs its own subaccount; some SAC guides use a separate subaccount for identity provisioning.

4.3 Recommended layout

Tier	Global account / subaccount	IAS tenant	Connected systems
Non-production	Enterprise global account → subaccount `joule-nonprod` (Cloud Foundry, supported region), Joule Preview entitlement where available	Test IAS tenant	Sandbox, development and quality systems of each product
Production	Same or separate global account → subaccount `joule-prod`	Productive IAS tenant	Productive systems of each product
Optional	Separate subaccounts for Joule for Consultants and for Joule Studio development	Same tier's IAS	As needed

5. People, access and readiness

5.1 Who you need

Role	Needed for	Access
SAP for Me administrator	SAP-managed provisioning, AI Unit activation, product packages	Cloud Administrator or Super Administrator in SAP for Me
BTP global account administrator	Entitlements, boosters, System Landscape, formations	Global Account Administrator
BTP subaccount administrator	Subscriptions, trust, role collections, destinations	Subaccount Administrator
Identity administrator	IAS applications, trusted domains, user attributes, IPS jobs, Entra ID federation	IAS administrator
Product administrator(s)	Product-side settings (communication arrangements, RBP, catalogs, OAuth clients)	Product admin roles
Basis administrator (Private Edition / on-premise)	SICF services, content exposure, certificate mapping, Cloud Connector	Basis authorisations
Network and security	Cloud Connector, allowlists, Content Security Policy, browser policies	Firewall and proxy owners
Business key users	Acceptance testing in each product	Test users with real roles

5.2 Readiness checklist

☐	Item	Owner	Ref
☐	Product licences and Joule Base / AI Units confirmed	Project lead	3
☐	Provisioning model chosen (SAP-managed or customer-managed)	Architect	2
☐	Enterprise BTP global account available	BTP admin	3
☐	IAS tenants identified per tier and already used by the products	Identity admin	4.2
☐	Every user has a Global User ID in IAS	Identity admin	7.6
☐	Supported region chosen for each subaccount	Architect	4.2
☐	Existing Work Zone, standard edition site identified for reuse	BTP admin	4.2
☐	Cloud Connector available (Private Edition / on-premise only)	Basis	8.2
☐	Test users per product with representative roles	Key users	11
☐	Data-protection review started	Security / DPO	14

6. Path A: SAP-managed provisioning

Use this path for new Joule deployments where SAP offers it for your products. SAP configures SAP BTP, Cloud Identity Services and the product integrations; you complete the product-side settings and assign users.

6.1 Steps

1. Confirm prerequisites: licensed SAP cloud product that supports Joule, in a supported region; Joule Base or AI Units purchased; all products on one IAS tenant per tier.
2. Sign in to **SAP for Me** with the Cloud Administrator or Super Administrator role.
3. Accept the SAP AI terms if prompted (some contracts use a zero-cost line item for this). VERIFY
4. Open **Portfolio & Products → Business AI** and find the Joule package. Choose **Ready for Activation**.
5. Open **Portfolio & Products → My Product Packages**, select the package and choose **Start Provisioning**.
6. Monitor progress on the **Resources** tab until provisioning completes.
7. Complete the product-side settings for each product (the "Product side" steps in section 8), for example role-based permissions in SuccessFactors or business catalogs in S/4HANA Cloud and IBP.
8. Assign users through the product's roles and the IAS groups SAP set up.
9. Run the tests in section 11.

Testing levels

SAP Community describes two SAP-managed test approaches: **Level 2** (recommended, customer-like end-to-end provisioning through SAP for Me) and **Level 0** (a quick background provisioning for preliminary validation when Level 2 is temporarily unavailable).

What you still own

User and role assignment, product-side configuration, Content Security Policy and trusted domains on the product side, testing, change management and AI Unit monitoring.

Extensions such as Joule Studio may have their own SAP-managed option (SAP Help: SAP-Managed Provisioning for Joule Studio).

7. Path B: customer-managed common foundation

These steps build the SAP BTP and identity foundation once per tier. Every product chapter in section 8 starts from here. SAP Help: *Customer-Managed Provisioning* and *Prerequisites for Customer-Managed Provisioning*.

7.1 Prepare the global account and subaccount

1. In the BTP cockpit, open the enterprise global account. Confirm the Joule entitlement is present under **Entitlements**.
2. Create or choose the subaccount (for example `joule-nonprod`) in a supported region, ideally the one that already hosts your Work Zone site.
3. Enable the **Cloud Foundry** environment and create a space.
4. Confirm the subaccount, the IAS tenant and Work Zone are in the same region.

7.2 Assign entitlements

Under **Entitlements → Entity Assignments**, add to the subaccount:

Service	Plan	Notes
Joule (`das-application`)	`foundation` in most guides; some newer guides show `standard` or DAS Standard / DAS Premium VERIFY	Joule Preview entitlement for non-production where offered (EU10 / EU30)
SAP Build Work Zone, standard edition	`foundation` (included in every global account); some SuccessFactors guides use `standard` VERIFY	Reuse an existing Work Zone subscription if you have one
SAP Cloud Identity Services	per your contract (guides cite `default`, `connectivity` or an additional-tenant plan) VERIFY	Only if the IAS tenant is not yet linked to the global account
Extensions (optional)	Document Grounding `data-manager`; SAP Build Process Automation `build_default`	See section 9

7.3 Subscribe to SAP Build Work Zone, standard edition

1. In **Instances and Subscriptions**, create the subscription to SAP Build Work Zone, standard edition (skip if it exists).
2. Assign the role collection `Launchpad_Admin` to the administrator.
3. Open the Work Zone site manager once to confirm it loads.

7.4 Establish trust with Identity Authentication (OpenID Connect)

1. In the subaccount, open **Security → Trust Configuration → Establish Trust** and select the IAS tenant that your SAP products use.
2. Confirm the trust type is **OpenID Connect**. Joule does not support SAML for this trust.
3. Where the guide for your product says so, disable the default identity provider for user logon so users always sign in through IAS.

7.5 Configure trusted domains

1. In IAS, add each product domain (for example the S/4HANA, SuccessFactors, Concur or Datasphere host) as a **trusted domain**.
2. In the BTP subaccount, add the product hosts under **Security → Settings → Trusted Domains**.
3. In each product, allow the Joule URL in its Content Security Policy or allowlist (for example **Manage Content Security Policy** in S/4HANA Cloud Public Edition).

7.6 Configure user attributes (Global User ID)

1. In IAS, open the application created for the subaccount trust.
2. Set the **subject name identifier** to the **Global User ID** from the Identity Directory.
3. Under assertion attributes, map `user_uuid` to the Global User ID. If `user_uuid` is not listed, add it as a custom attribute sourced from the Identity Directory.
4. Set the Work Zone application's subject name identifier to the Global User ID as well, so the navigation service can match users.
5. Check that every user exists in the IAS user store with a Global User ID; Joule cannot recognise users without one.

7.7 Register product systems

Products that need registration (for example S/4HANA Cloud Public Edition and IBP) must appear in the global account under **System Landscape → Systems** before the booster can integrate them. Many SAP cloud products appear automatically; others need a registration token configured in the product. Follow the product chapter in section 8. VERIFY

7.8 Run the Joule booster

1. In the global account, open **Boosters**, search for **Setting Up Joule** (shown as "Joule" in some releases) and choose **Start**.
2. Let the booster check entitlements, authorisations and the IAS tenant. Fix anything it reports (see section 13: missing Work Zone registration, missing IAS tenant).
3. Select the subaccount, the integration scenarios, and the SAP systems to integrate. Give the formation a name, for example `joule-nonprod`.
4. Finish the booster. It creates the Joule subscription, destinations and the formation, and (for S/4HANA Cloud Public Edition) communication arrangements in the background.
5. Check **System Landscape → Formations**: the formation should show status **Ready** and list your systems.
6. To add another product later, open the formation and use **Include Systems**, or run the booster again for that system.

> **The booster does not create role collections**
> Create them yourself in 7.9, or users will not see Joule even though the setup succeeded.

7.9 Create role collections and assign users

Role collection (suggested name)	Roles from application `das-application`	Assign to
`JOULE_END_USER`	`end_user`	Every Joule user (map to an IAS group)
`JOULE_ADMIN`	`tenant_admin`, `analytics_admin` and other admin roles listed in the subaccount	Joule administrators

Role collection (suggested name)	Roles from application `das-application`	Assign to
`JOULE_EXTENSIBILITY` (optional)	`extensibility_developer`	Joule Studio developers

1. In the subaccount, open **Security → Role Collections → Create**.
2. Add the roles above. Role names can differ slightly per data centre; pick them from the `das-application` list. VERIFY
3. Map each role collection to an IAS user group (**Trust Configuration → Role Collection Mappings**) or assign users directly under **Security → Users**.
4. Prefer least privilege: do not hand out admin role collections to test end users.

7.10 Configure Identity Provisioning

1. In the Identity Provisioning Service, create a source system (IAS or the product) and a target system **SAP Build Work Zone** for the subaccount.
2. Map users, groups and roles; include `userUuid` (Global User ID).
3. Run the job once, then schedule it (for example hourly) so role changes reach Work Zone.

7.11 Foundation checklist

☐	Item	Owner	Ref
☐	Subaccount in supported region, Cloud Foundry enabled	BTP admin	7.1
☐	Joule, Work Zone and Cloud Identity Services entitlements assigned	BTP admin	7.2
☐	Work Zone, standard edition subscribed; Launchpad_Admin assigned	BTP admin	7.3
☐	OIDC trust to the shared IAS tenant	Identity admin	7.4
☐	Trusted domains in IAS, BTP and product CSP	Identity / product admin	7.5
☐	Global User ID as subject name identifier; user_uuid mapped	Identity admin	7.6
☐	Product systems registered in System Landscape	BTP admin	7.7
☐	Booster finished; formation status Ready	BTP admin	7.8
☐	Role collections created and mapped to IAS groups	BTP admin	7.9
☐	IPS job to Work Zone scheduled	Identity admin	7.10

8. Product integrations

Each chapter assumes the foundation in section 7 (or SAP-managed provisioning in section 6). SAP Help hosts one integration page per product under *Integrating Joule with SAP*.

8.1 SAP S/4HANA Cloud Public Edition

Prerequisites

- S/4HANA Cloud Public Edition tenant using the same IAS tenant as the BTP subaccount trust.
- Administrator with access to communication management and Content Security Policy apps; `Launchpad_Admin` in BTP.
- Only one S/4HANA Cloud Public Edition tenant can currently connect to a Joule tenant (stated for Joule for Developers). VERIFY

Steps

1. Register the S/4HANA Cloud system in **System Landscape** (if it is not already listed) and complete the registration in S/4HANA. VERIFY
2. Run the booster (7.8) and select the S/4HANA Cloud system. In the background it creates communication arrangements `SAP_COM_0882` and `SAP_COM_0647`.
3. In S/4HANA, open communication arrangement `SAP_COM_0647` and set the role exposure to **ALL**. If you keep **SELECTED**, maintain the filter `s4hana.cloud.roles.filter – cFLGExposure eq true`.
4. Create and activate `SAP_COM_0193` manually so users and roles are provisioned to Work Zone (with IPS, 7.10).
5. In **Manage Content Security Policy**, add the Joule application URL to the trusted sites.
6. Add the S/4HANA tenant URL to the BTP subaccount trusted domains.
7. In Work Zone, refresh the S/4HANA content provider so apps and roles are available to the navigation service.
8. Make sure test users have business roles whose apps are exposed; for Joule for Developers add catalog `SAP_A4C_BC_DEV_AIQ_PC` (allow up to 24 hours).
9. Sign out and in again (or use a private window), open the Fiori launchpad and check the Joule icon.

> **Source**
> SAP Help: Integration with SAP S/4HANA Cloud Public Edition; SAP Community: Joule in SAP S/4HANA Cloud Public Edition – Setup Guide.

8.2 SAP S/4HANA Cloud Private Edition (RISE)

Prerequisites

- RISE with SAP contract; S/4HANA release **2021 or later**, with **2023 FPS01** recommended.
- SAP Cloud Connector connected to the Joule subaccount; principal propagation configured.
- Fiori launchpad users authenticate through the shared IAS tenant.
- SAP Note `3639572` (Enabling Joule with SAP S/4HANA Cloud Private Edition) read and applied. VERIFY

Steps (SAP Help order)

1. Establish trust between the subaccount and IAS (7.4).
2. Configure trusted domains in BTP and IAS, including the S/4HANA launchpad host (7.5).
3. Configure user attributes so S/4HANA users carry the Global User ID (7.6).

4. Expose Fiori launchpad content: activate the content-exposure HTTP services in transaction SICF, either `/sap/bc/ui2/cdm3` (version 1) or `/sap/bc/http/ui2/flp_content_exposure/entities` (version 2, from 2023 FPS01). Then run `/UI2/CDM3_EXP_SCOPE` to select the roles to expose.
5. Check exposure logs in `/UI2/CDM3_EXP_SCOPE` (View Log) or transaction SLG1, object `/UI2/BE`, sub-object `/UI2/CDM3`.
6. In Cloud Connector, expose the S/4HANA host with the content-exposure path and all sub-paths, plus the OData paths Joule scenarios use. Configure principal propagation (trust to the subaccount, system certificate, certificate mapping in S/4HANA).
7. Create destinations: a design-time destination (content exposure, basic authentication) and a runtime destination (principal propagation) for Work Zone.
8. In Work Zone, open **Channel Manager → New** and add the S/4HANA system as a content provider; fetch content.
9. Run the booster (7.8) for the Private Edition scenario and check the destinations it creates.
10. Configure IPS so S/4HANA roles and users reach Work Zone (7.10).
11. Enable Joule in the S/4HANA Fiori launchpad as described in SAP Note `3639572`. VERIFY
12. Schedule a regular content-update job so launchpad content stays in sync.
13. Test with real business roles (section 11).

> **Automation and multi-system**
> The **Cloud Integration Automation service** can drive much of this provisioning for Private Edition.
> SAP published multi-system Joule support for Private Edition in 2026, for landscapes with more than one S/4HANA system. VERIFY
> Related notes: `3523238` (semantic objects and actions list), `3641945` (Joule icon displayed wrongly in the launchpad).

8.3 SAP S/4HANA on-premise and SAP ECC

Until 2026 SAP did not offer native Joule for on-premise S/4HANA or ECC; the documented route was a hybrid setup through BTP and Cloud Connector, mainly for proofs of concept. At SAP Sapphire 2026, SAP announced that **Joule Work** and a share of Joule assistants and agents would work in hybrid landscapes with on-premise S/4HANA and ECC, tied to a cloud commitment (sources mention RISE and the Max Success Plan). VERIFY

1. Ask SAP whether your contract qualifies and which release levels are supported.
2. If it does, follow the Private Edition pattern (8.2): IAS trust, Global User ID, content exposure, Cloud Connector with principal propagation, Work Zone content provider, booster.
3. Treat SAP Note `3632703` (cited for the earlier on-premise position) and SAP's current announcement materials as the authority. VERIFY

8.4 SAP SuccessFactors

Prerequisites

- Valid SuccessFactors licence on a data centre that supports Joule.
- SuccessFactors authenticating through SAP Cloud Identity Services (usually delivered with the SuccessFactors tenant), with IPS in use.
- SAP Build Work Zone, standard edition / SAP Start available.
- SAP Discovery Center mission **Activate Joule for SAP SuccessFactors** for guided steps.

Steps

1. Confirm Cloud Identity Services is active and establish trust between it and the subaccount (7.4). SAP states this is required for the updated booster.
2. In IAS, add the SuccessFactors domain as a trusted domain.
3. Run the booster (7.8) and select the SuccessFactors system.
4. In Cloud Identity Services, add a source system, import the Joule JSON file for SuccessFactors, and rename the system to your SuccessFactors company ID. VERIFY
5. In SuccessFactors **Admin Center → Manage Permission Roles**, grant **General User Permission → Access to Joule** to the roles that should use Joule, and assign those roles to groups.
6. Provision the groups to Work Zone with IPS (7.10).
7. Sign in as a test user and check the Joule button. If it is missing for some users, see KBA 3685621.

> **Extension agents**
> Custom agents that read SuccessFactors data should propagate the user's identity so role-based permissions apply, rather than running on technical-user credentials.

8.5 SAP Concur

1. Configure trust to IAS and trusted domains in BTP and IAS (7.4, 7.5).
2. Configure user attributes from the Identity Directory (7.6).
3. Run the booster and select the Concur system. For an existing formation, use **Include Systems**.
4. Check the formation status is **Ready** and the Joule tenant appears in the systems landscape.
5. Add the Concur domain under the security settings as the post-integration step.
6. Assign users and test.

8.6 SAP Ariba

1. Make sure you have the Ariba Organization Admin role and know your realm's data centre.
2. Create the Ariba applications that generate the OAuth client ID and secret.
3. Configure single sign-on through SAP Cloud Identity Services.
4. Run the booster and choose Ariba Buyer and Ariba Sourcing systems.
5. Assign users and test.

> **Source**
> SAP Community: Joule Integration with SAP Ariba – Setup Guide. Confirm the current steps on the SAP Help Ariba integration page. VERIFY

8.7 SAP Analytics Cloud

- SAP Build Work Zone is required (since Joule 2506): it provisions and syncs SAC users and roles; SAC roles and teams must sync to Work Zone as content.
- SAC authenticates with SAP Cloud Identity Services; keep at least **three free OAuth client slots**.
- Only one SAC tenant per Joule formation.
- Users need **Read** on the Joule privilege; enable **Just Ask** under the Conversational Analytics settings (check under System Administration → Default Appearance).
- For analytical insights, the user's SAC role needs read access to the content.

8.8 SAP Datasphere

- Cloud Foundry subaccount; subscription-based, CPEA or SAP-managed Datasphere tenant with at least **one free OAuth client slot**.
- Users need role **DW Joule Consumer** (or a role with the Joule privilege); the tenant administrator activates Joule.
- The subaccount XSUAA subject name identifier must match the Datasphere identity provider setting.
- Add Datasphere trusted domains to the Joule IAS; synchronise users, roles and permissions to Work Zone with IPS.

8.9 SAP Integrated Business Planning

1. Register the IBP system with SAP BTP.
2. Run the booster with the IBP system and complete the IAS trust.
3. Add business catalog **User Interface – Joule Plugin** (`SAP_CORE_BC_UI_JOU_PC`) to the users' business roles.
4. Test with planners' roles.

8.10 Other SAP products

Product	Where to start
SAP Signavio solutions	SAP Help: Integration with SAP Signavio Solutions
SAP Fieldglass	SAP Help: Integrating with SAP Fieldglass
SAP Integrated Product Development	SAP Community: Joule Setup for SAP Integrated Product Development
SAP Sales Cloud / Service Cloud Version 2 and other CX	SAP Help integration page for the product VERIFY
SAP Integration Suite, SAP Build	SAP Help integration pages; Joule features are often enabled inside the product VERIFY
Unified setup across several products	SAP Community: Joule end-to-end setup guide for all lines of business (unified approach); SAP Learning: Setting Up a Unified Joule Instance

9. Extending Joule

9.1 Document grounding (answers from your own documents)

Grounding lets Joule answer from company documents. Documents are turned into vector embeddings by the Document Grounding service on SAP BTP and stored in SAP HANA Cloud Vector Engine.

1. In the Joule subaccount (or as SAP's guide directs), add the **Document Grounding** entitlement (plan `data-manager`), create the instance and a service key.
2. For Microsoft SharePoint: register an app in Microsoft Entra ID, grant it **Sites.Selected** (least privilege, preferred over Sites.Read.All), grant it access to the specific sites, and record the client ID and secret.
3. Create a pipeline for each source: SharePoint, SAP Document Management Service (via a BTP destination) or Google Drive.
4. Run the pipeline and check its status; then enable grounding for Joule as the guide describes.
5. Optionally add metadata ("contextualization") so answers can be filtered, and use the Cloud Integration Automation service to automate setup.
6. Test with questions whose answers exist only in the documents.

9.2 Microsoft 365 Copilot and Microsoft Teams

1. Federate SAP Cloud Identity Services with Microsoft Entra ID using **OpenID Connect** (SAML is only suitable when Entra is used purely for login).
2. Configure user mapping so Entra and IAS users match.
3. Register the SAP Joule application in Entra ID; grant admin consent for the SAP Copilot API access app, then under the federation app's **API permissions** add `svc-ai-joule-copilot-api-access` with delegated `access_as_user` and grant consent. VERIFY
4. In BTP System Landscape, add a system of type **Microsoft 365 Copilot for SAP Joule** and include it in the Joule formation.
5. Enable the Joule agent in the Microsoft 365 admin centre or Teams admin centre.
6. Use a separate Entra ID tenant per Joule environment (a single Entra tenant cannot serve several Joule environments).
7. Test both directions: Copilot/Teams passing requests to Joule, and Joule passing requests to Copilot.

9.3 Joule Studio (custom skills and agents)

1. Entitlements: SAP Build Process Automation with plan **`build_default`** (tenants on plan `standard` are not compatible), Joule, and an SAP Cloud Identity Services tenant.
2. Configure OIDC trust to IAS and disable the default identity provider for login.
3. Run the Joule booster from the global account if Joule is not yet set up.
4. Create role collections: a Joule admin collection with the Joule roles, and `ProcessAutomationAdmin` (plus developer roles) for builders; assign to your IAS users or groups.
5. Configure destinations to back-end systems the skills will call.
6. Build a skill or agent in SAP Build (Joule Studio), test it, and deploy it to the Joule tenant; use the SAP Build lifecycle to promote from test to production.
7. Follow the SAP Discovery Center mission **Setup Joule Studio for a BTP Enterprise Account**, or SAP-managed provisioning for Joule Studio where available.

9.4 Joule Agents

- SAP-delivered Joule Agents are premium capabilities that consume AI Units.
- Administrators grant users permission to specific agents through the product's roles (for example SuccessFactors agents through role-based permissions).
- Plan consumption in advance and monitor it in SAP for Me and the Joule Analytics Center (12.2).

10. Joule for Consultants and Joule for Developers

10.1 SAP Joule for Consultants

- Admins needed: BTP global account administrator, SAP Cloud Identity Services administrator, and Cloud Administrator or Super Administrator in SAP for Me.
- Use a **dedicated subaccount**; integrating with another Joule instance in the same subaccount does not work.
- Supported regions cited: EU10, EU11, EU20, US10, AP10, JP10. VERIFY

1. Assign entitlements: Joule (`foundation`) and **SAP Consulting Capability** (`standard ga`).
2. Activate the service from the subscription; check **Instances and Subscriptions**.
3. Create a role collection with `end_user` from `das-application`, for example `JOULE_FOR_CONSULTANTS_END_USER-<subaccount>`.
4. Assign users through an IAS group mapped to the role collection, or directly.
5. Open the console from **Instances and Subscriptions**. Optionally add custom knowledge grounding.

10.2 SAP Joule for Developers (ABAP AI capabilities)

System	Setup
S/4HANA Cloud Public Edition / BTP ABAP environment	Assign users a business role containing catalog `SAP_A4C_BC_DEV_AIQ_PC`; allow up to 24 hours.
S/4HANA Cloud Private Edition 2025 (newer posts: 2021 and higher)	Subscribe to the service in BTP; generate a client certificate; configure certificate mapping in S/4HANA (Maintain Certificate Mapping app, IAS group required); assign authorisation object `S_AIQADTLO`; allowlist the regional proxy URL. SAP Note `3558794` covers known issues. VERIFY

11. Testing and acceptance

Test in the non-production tier with real business roles, then repeat the core tests in production before go-live.

ID	Test	How	Expected result
T01	Single sign-on	Open the product as a test user in a private window and launch Joule	Joule opens without a second login
T02	Navigation	"Open Manage Purchase Orders" (or a common app in the product)	Joule offers the app the user is allowed to open
T03	Informational	"What is a credit memo?" / "How do I request leave?"	Grounded answer from SAP help content
T04	Transactional	"Show my open purchase orders" / "Show my team's time off"	Live data from the product, limited to the user's permissions
T05	Authorisation (negative)	Repeat T04 with a user lacking the business role	Joule declines or shows no data
T06	Cross-product	Ask something that needs a second integrated product	Joule routes to the right product
T07	Document grounding	Ask about content only in your grounded documents	Answer cites the document
T08	Microsoft 365	Ask Joule from Teams / Copilot, and Copilot from Joule	Requests pass both ways
T09	Custom skill or agent	Trigger the Joule Studio skill	Skill runs with the user's identity
T10	Usage analytics	Open the Joule Analytics Center after tests	Test sessions appear by product and scenario
T11	Mobile / other clients	Use Joule from the supported mobile or desktop client if in scope	Same behaviour as browser

Exit criteria

All tests pass for every product in scope, no admin roles are assigned to test end users, analytics show the sessions, and open issues are logged with SAP (component `CA-JOULE`) where needed.

12. Operating Joule

12.1 Administration

- **Joule Admin Center:** `<joule-tenant-url>/admin`, role `tenant_admin`.
- **Role management:** one shared role collection for all Joule roles, or separate ones for finer control; review quarterly.
- **User provisioning:** keep IPS jobs scheduled; leavers must be removed in IAS so Joule access ends.
- **Release cadence:** Joule updates frequently; validate new capabilities on the Joule Preview (non-production) instance before they reach production.

12.2 Usage and cost monitoring

- **Joule Analytics Center:** `<joule-tenant-url>/analytics`, role `analytics_admin`. Shows product usage, scenario usage, interaction-type usage and client-type usage. SAP describes it as the dedicated source of Joule usage; usage is not shown in SAP for Me.
- **AI Units:** track balance and consumption in SAP for Me; agree alert thresholds with Finance.

12.3 Support

- Open SAP incidents under component `CA-JOULE` (product-specific components may also apply).
- Keep a runbook of your formation name, subaccounts, IAS tenants, destinations and role collections for each tier.

13. Troubleshooting

Symptom	Likely cause	Fix
Booster stops at the Work Zone check	SAP Build Work Zone not subscribed or not registered in the subaccount	Subscribe to Work Zone, standard edition in the Joule subaccount; see KBA `3721340`
Booster reports no Identity Authentication tenant	The global account has no IAS tenant linked	Link or create the Cloud Identity Services tenant; see KBA `3675105`
Joule icon missing for everyone	Role collection not assigned; Content Security Policy or trusted domain missing; Joule not enabled in the launchpad; browser cache	Check 7.5 and 7.9; for S/4HANA Public re-check "Update Content Security Policy Settings"; try a private window and clear cache
Joule works for admins, not for users	Users lack the `end_user` role collection or the IAS group mapping	Map the role collection to the users' IAS group (7.9)
Joule button missing for some SuccessFactors users	Missing General User Permission → Access to Joule; roles not synced; browser or network policy blocks URLs	Follow KBA `3685621`: permissions, sync, third-party cookies, allowlist SuccessFactors, IAS and Joule URLs
Login loop or session breaks after Joule opens	Different IAS tenants, SAML trust, or subject name identifier not the Global User ID in Joule or Work Zone	Use one IAS tenant, OIDC trust, Global User ID everywhere (7.4, 7.6)
"App not found" or wrong app on navigation	Work Zone content not synced, roles not provisioned, or (Private Edition) content not exposed	Refresh the content provider, run IPS, check `/UI2/CDM3_EXP_SCOPE` and SLG1
Private Edition content provider fails	SICF service inactive, Cloud Connector path missing, wrong destination	Activate the content-exposure service, expose the path with sub-paths, check the design-time destination (8.2)
Joule icon displayed wrongly in the Private Edition launchpad	Known issue	SAP Note `3641945`
SAC "Just Ask" not offered in Joule	Privilege or feature not enabled, no free OAuth clients, or second SAC tenant in the formation	Follow 8.7
Grounding returns nothing	Pipeline failed or not run; SharePoint consent missing	Check pipeline status; grant Sites.Selected access to the site (9.1)
Teams / Copilot cannot reach Joule	Entra–IAS trust on SAML, consent missing, Entra tenant reused across environments	Use OIDC, grant `svc-ai-joule-copilot-api-access`, one Entra tenant per environment (9.2)
Joule not supported in ADT (Private Edition)	Known issue for ABAP AI capabilities	SAP Note `3558794`
Unexpected AI Unit consumption	Agents or premium features used more than planned	Review the Joule Analytics Center and SAP for Me; adjust agent permissions

> **Escalation**
> If the fixes above do not help, open an incident under `CA-JOULE` with the formation name, subaccount ID, IAS tenant, product system, user, time of the attempt and a HAR file from the browser.

14. Security and data protection

- Joule acts with the signed-in user's identity; product authorisations decide what data it can show or change. Keep role design and segregation of duties current.
- SAP states it does not share customer data with third-party model providers to train their models, and that prompts sent to third-party hosted models are processed, not retained. Model routing is SAP-managed through the generative AI hub in SAP AI Core. VERIFY
- Confirm data residency and retention for your regions in the Joule security guide and SAP's data-protection documentation before production. VERIFY
- Use least-privilege access for grounding sources (for example SharePoint Sites.Selected) and review which documents are indexed.
- For custom agents, propagate user identity rather than using technical users, so permissions apply.
- Include Joule in your AI governance: acceptable use, human review of agent actions, audit of admin role assignments, and incident handling.

14.1 Security checklist

☐	Item	Owner	Ref
☐	Data-protection impact assessment completed	DPO	14
☐	Residency and retention confirmed for chosen regions	Security	14
☐	Admin role collections limited to named administrators	BTP admin	7.9
☐	Grounding sources reviewed; least-privilege access	Content owner	9.1
☐	Custom skills use principal propagation / user identity	Developers	9.3
☐	User communication and acceptable-use guidance published	Change lead	15

15. Master go-live checklist

Phase 1: Prepare

☐	Item	Owner	Ref
☐	Licences, Joule Base and AI Units confirmed	Project lead	3
☐	Provisioning model and landscape (tiers, regions, IAS tenants) approved	Architect	2, 4
☐	People and access in place	Project lead	5
☐	Global User ID present for all users	Identity admin	7.6

Phase 2: Build (non-production)

☐	Item	Owner	Ref
☐	SAP-managed provisioning complete, or customer-managed foundation complete	BTP admin	6 / 7
☐	Each product integrated and product-side settings done	Product admins	8
☐	Extensions configured (grounding, Microsoft 365, Joule Studio) if in scope	BTP admin	9
☐	Tests T01–T11 passed	Key users	11

Phase 3: Go live (production)

☐	Item	Owner	Ref
☐	Production tier built with the same steps	BTP admin	6 / 7 / 8
☐	Core tests repeated in production	Key users	11
☐	Role collections mapped to production IAS groups	Identity admin	7.9
☐	Users informed; quick-start guide and support route published	Change lead	12.3
☐	Analytics and AI Unit monitoring switched on	Joule admin	12.2

Phase 4: Operate

☐	Item	Owner	Ref
☐	Monthly review of usage, AI Units and incidents	Joule admin	12
☐	New Joule releases validated on Joule Preview first	Joule admin	12.1
☐	Quarterly role and access review	Security	14

Appendix A. Glossary

Term	Meaning
AI Units	SAP's consumption currency for premium AI capabilities such as Joule Agents.
Booster	Guided automation in the BTP cockpit; "Setting Up Joule" configures Joule in a subaccount.
CIAS	Cloud Integration Automation service: automates multi-step provisioning (Private Edition, grounding).
Content exposure (CDM3)	How S/4HANA Private Edition / on-premise publishes launchpad content to Work Zone.
Formation	A BTP System Landscape grouping of Joule and the systems it integrates.
Global User ID	The person-level identifier (`user_uuid`) that must match across IAS and all products.
IAS / IPS	Identity Authentication and Identity Provisioning, the two parts of SAP Cloud Identity Services.
Joule Base / Premium	Included Joule capabilities vs AI-Unit-based premium capabilities.
Joule Preview	SAP's preview landscape for non-production Joule tenants.
Joule Studio	The SAP Build capability for custom Joule skills and agents.
Principal propagation	Passing the signed-in user's identity from BTP to a private back end through Cloud Connector.
Work Zone, standard edition	SAP Build Work Zone; Joule uses its navigation service to find apps.

Appendix B. SAP Notes and Knowledge Base Articles

Open these in SAP for Me (support.sap.com) before setup. Titles are as cited in research; confirm the current version.

Number	Topic
2791410	SAP Cloud Identity Services integration: required configuration and considerations (cited for SuccessFactors)
3523238	Semantic objects and actions supported by Joule (spreadsheet)
3558794	AI functionality in S/4HANA Cloud Private Edition (Joule for Developers / ADT)
3632703	Joule and on-premise S/4HANA (earlier position) VERIFY
3639572	Enabling Joule with SAP S/4HANA Cloud Private Edition
3641945	Joule icon displayed wrongly in the Fiori launchpad of S/4HANA Cloud Private Edition
3675105	Joule booster: global account without an Identity Authentication tenant
3685621	Joule button not appearing on SuccessFactors
3721340	Joule booster: SAP Build Work Zone registration missing
3721491	Joule availability question (on-premise) VERIFY
3729978	Joule setup (SuccessFactors context) VERIFY

Appendix C. Key sources

The full list with confidence ratings is in `research-notes.md`. Start with these SAP pages:

- Joule Integration Guide (SAP Help, PDF): https://help.sap.com/doc/de3af3c0f81642dbaa4d36172ed57a72/CLOUD/en-US/79bfc83ab386450c8cd9c7937ce26a3a.pdf
- What is Joule (SAP Help, PDF, 2026-10-20): https://help.sap.com/doc/75f5965fbb514fbcb89f1e47c094ea3f/CLOUD/en-US/9d8c0ef2443b49f993a27de8f38f83af.pdf
- Joule Onboarding Guide: https://help.sap.com/docs/joule/serviceguide/onboarding-joule
- Customer-Managed Provisioning: https://help.sap.com/docs/joule/integrating-joule-with-sap/initial-setup
- Prerequisites for Customer-Managed Provisioning: https://help.sap.com/docs/joule/integrating-joule-with-sap/prerequisites
- Integration with S/4HANA Cloud Public Edition: https://help.sap.com/docs/joule/integrating-joule-with-sap/integration-with-sap-s-4hana-cloud-public-edition
- Integration with S/4HANA Cloud Private Edition: https://help.sap.com/docs/joule/integrating-joule-with-sap/integration-with-sap-s-4hana-cloud-private-edition
- Identity and Access Management for Joule: https://help.sap.com/docs/cloud-identity-services/cloud-identity-services/identity-and-access-management-for-joule
- Joule Landscape Recommendation (Architecture Center): https://architecture.learning.sap.com/docs/ref-arch/4c80fa
- Setting up Joule across your system landscapes (SAP Learning): https://learning.sap.com/courses/setting-up-joule-across-your-organizational-system-landscapes
- Joule end-to-end setup guide for all lines of business (SAP Community): https://community.sap.com/t5/technology-blog-posts-by-sap/joule-end-to-end-setup-guide-for-all-line-of-business-unified-approach/ba-p/13964602
- Managing Joule on SAP BTP (SAP Community): https://community.sap.com/t5/technology-blog-posts-by-sap/managing-joule-on-sap-btp-roles-analytics-administration-and/ba-p/14325933
- SAP Joule and Microsoft Copilot integration (Microsoft Learn): https://learn.microsoft.com/en-us/azure/sap/microsoft-ai/joule/joule-copilot-overview

Appendix D. Open items to verify

These details came from a single or non-SAP source, or sources disagreed. Close each one against SAP Help, SAP for Me or your SAP account team.

#	Item	Where used
1	Exact plan names for Joule (`foundation` vs `standard` / DAS Standard / Premium), Work Zone and Cloud Identity Services	7.2
2	SKU numbers and current AI Unit consumption rates	3
3	Supported regions for each Joule product and for Joule Preview	4.2, 10.1
4	Role names inside `das-application` for your data centre	7.9
5	S/4HANA Cloud Public Edition system registration steps	7.7, 8.1
6	Private Edition: Joule launchpad enablement per Note 3639572; multi-system support	8.2
7	On-premise / ECC eligibility, scope and dates after Sapphire 2026	8.3
8	SuccessFactors JSON source-system import in Cloud Identity Services	8.4
9	Ariba, CX, Integration Suite, Fieldglass, Signavio detailed steps	8.6, 8.10
10	Entra ID permission names for the Microsoft 365 integration	9.2
11	Joule for Developers on Private Edition: releases and authorisations	10.2
12	Data residency and retention statements	14